

Exploring U.S. Emergency Department Visits: NHAMCS-ED 2015

MGT 634 Empirical Strategy with AI · Homework 1 · Group: Michal Kozuchowski, Laila Lapins, Nick Giamalis, Raymond Chang, Sean Weller · September 2026

Bottom line. U.S. EDs handled an estimated **136.9 million visits** in 2015. Demand is predictable: it plateaus from 10am to 8pm and is highest on Mondays. Payment is dominated by public payers (**Medicaid 34.9%, Medicare 19.9%** of visits with a known payer), and **nearly half of visits (47.6%)** involve a chronic condition. Triage sorts patients sharply by outcome (a third of the most urgent are admitted vs 2-3% of the least urgent), but median waits are almost identical (18-20 minutes) for every level except the most urgent.

1. Data, sample design and cleaning (Q1-Q2)

What the data are. The NCHS public-use file has **21,061 rows and 1,031 columns**. Each row is one sampled ED visit (one Patient Record Form), not a patient: the same person could appear twice. Hospital code + patient code uniquely identify every row (0 duplicates), and the visits come from **248 EDs** (median 94 sampled visits each). Summing the survey weight PATWT gives **136,943,181 estimated U.S. ED visits** in 2015, exactly the documented total and well over 100 million. Weights are very unequal (117 to 42,002, median 5,232; the top 10% of rows carry 28.7% of weighted visits), and the sample over-represents the Northeast (20.4% of rows vs 17.3% of visits) while under-representing the South (33.3% vs 37.8%). We therefore weight every total, share and rate by PATWT.

Missing data are hidden as codes. The standard pandas missing-value check (isna) reports NaN in 600 columns, but these are mostly empty drug-category text slots. Real missing answers are stored as negative numbers: **-9 = blank, -8 = unknown, -7 = not applicable** (codebook p.37 onward). Counting those codes, 118 variables are more than 5% missing and 56 are more than 20% missing. For our variables: unimputed ethnicity 24.2% (so we use the imputed ETHIM, 0% missing), pain scale 29.5%, triage level 22.2%, wait time 15.2% blank (+3.4% not seen by a provider), payer 8.7%, length of visit 7.0%. INTENT15 (75% blank) and HDSTAT (91% "not applicable") are structurally missing: they are only asked for injury visits and admitted patients. We confirmed this reading of the codes by reproducing NCHS's published nonresponse rates (Table 1).

Missingness is not random. Wait time is missing for 7-12% of triaged visits but 38-45% of visits with no triage recorded, and for 9.8% of Midwest vs 19.7% of Northeast visits. This is a missing-at-random pattern tied to observable ED processes. Wait-time statistics therefore describe EDs that record wait times, and may understate waits in less-structured settings.

Cleaning rules. (1) Convert every -9/-8/-7 to missing before computing anything. (2) Recode yes/no items to 0/1. (3) Convert ARRTIME from military time to an arrival hour (1.3% blank). (4) Handle long tails rather than delete them: length of visit reaches 5,732 minutes (95 hours) and waits 1,305 minutes, so we report medians and winsorize means at the 99.5th percentile (98 and 86 visits capped). (5) Test the weighted mean, median and correlation functions on data with known answers, and re-check every published NCHS figure we could find. A final clean run passes all 20 of 20 verification checks.

Table 1. Our estimates reproduce every published NCHS figure we checked

Check	Our result	NCHS codebook
Records	21,061	21,061 (p.115)
Weighted U.S. ED visits	136,943,181	136,943,181 (p.28)
Female share (weighted)	55.44%	55.436% (p.115)
Hispanic share (weighted)	16.49%	16.493% (p.115)
Medicaid as primary payer (weighted)	31.15%	31.152% (p.116)
Visits with no medications (weighted)	20.94%	20.943% (p.117)
Weighted drug mentions	340,550,921	340,550,921 (p.118)
BUN/creatinine test ordered	6.67%	6.7% (p.4)
Length of visit / wait time missing	7.0% / 15.2%	7.0% / 15.7% (p.15)

2. Who comes to the ED, and how long they stay (Q3)

Table 2. Descriptive statistics, unweighted vs weighted

Statistic	Unweighted	Weighted
Age: mean / median (years; 93 = top-code "93+")	37.6 / 34	37.0 / 34
Female / Hispanic (imputed ETHIM)	55.1% / 15.9%	55.4% / 16.5%
Wait to see a provider: median (n = 17,153)	19 min	18 min
Wait: mean, winsorized 99.5%	39.5 min	38.7 min
Length of visit: mean raw / winsorized (n = 19,581)	221.9 / 216.5 min	213.7 / 209.8 min
Length of visit: median	154 min	154 min

Patient flow. The typical patient is a young adult (median age 34), and 55% of visits are by women. Half of patients see a provider within **18 minutes**, but the mean wait (39 minutes) is twice the median. Length of visit is similarly skewed (median **2.6 hours**, mean 3.5 hours): a minority of very long waits and stays (for example, patients boarding while they wait for an inpatient bed) consumes a disproportionate share of bed-hours. By Little's law (average patients in the ED = arrival rate x time spent there), 375,000 visits/day x 3.56 hours means about **55,700 patients are in U.S. EDs at any moment**, roughly 12 per ED across the country's 4,820 EDs (codebook p.120). ED capacity is driven as much by throughput time as by arrivals.

Effect of weighting. Weighting moves the headline numbers only slightly: mean age falls 0.5 years, mean length of visit falls 8 minutes, and children rise from 18.6% to 19.8% of visits. That's because the weights shift toward the under-sampled South and toward children, who have shorter visits. Medians do not move. So the sample is broadly representative, but unweighted counts are meaningless for volume (21,061 vs 136.9 million), and small tilts compound in subgroup analysis. We therefore weight everything.

3. When demand arrives (Q4)

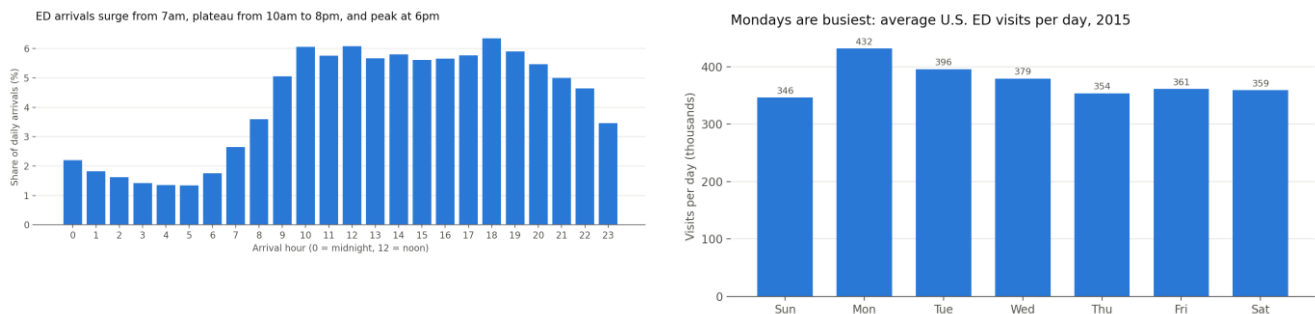


Figure 1. Weighted ED arrivals by hour of day (left) and average visits per day by weekday (right)

Busiest days and times. Because 2015 had 53 Thursdays, we compare weekdays as average visits per day. **Monday is busiest (432,000 visits/day)** and Sunday quietest (346,000), a 25% gap. Volume declines from Monday through Thursday. Arrivals bottom out between 3am and 6am (1.3-1.4% of daily arrivals per hour), climb steeply from 7am, and **plateau from 10am to 8pm**, peaking at **6pm (6.3%)**. The peak hour receives 4.7 times as many arrivals as the quietest, 69% of patients arrive between 10am and 10pm, and the single busiest slot is Monday 9-10am (about 28,500 visits nationally).

Staffing implications. Staff to a long daytime plateau rather than a single peak. Stagger shift starts through the 7-10am ramp, and keep full coverage until roughly midnight: a patient arriving at the 6pm peak stays a median 2.6 hours, so occupancy peaks later than arrivals do. Roster more heavily on Mondays and Tuesdays, consistent with pent-up weekend demand while doctors' offices are closed. Overnight volume runs at about a quarter of peak, which suits lean staffing and on-call coverage.

4. Payer mix and financial exposure (Q5)

PAYTPER assigns each visit one primary expected payer using an NCHS hierarchy. Across all visits (weighted), **Medicaid/CHIP pays for 31.2%, private insurance 27.6%, Medicare 17.7% and self-pay 9.0%**; for 10.8% the payer is blank or unknown. Among the 89.2% of visits with a known payer: **Medicaid 34.9%, private 30.9%, Medicare 19.9%, self-pay 10.1%**. Public payers together cover 54.8%, and self-pay plus charity care cover 11.0%. Payer mix is driven by age (Figure 2): Medicaid pays for 65% of child visits, Medicare for 78-87% of visits by patients 65 and over, and self-pay peaks at 17.5% among 25-44-year-olds.

Financial performance. More than half of ED revenue comes from public payers, which typically reimburse below private rates, and about 1 in 9 visits is uninsured or charity care. Federal law (EMTALA) requires EDs to screen and stabilize everyone regardless of ability to pay, so hospitals cannot turn this demand away. Privately insured visits (under a third) cross-subsidize the rest. The 10.8% of visits with an unknown payer point to a documentation and revenue-cycle gap. Young adults are the main bad-debt exposure, and an aging population will shift mix further toward Medicare.

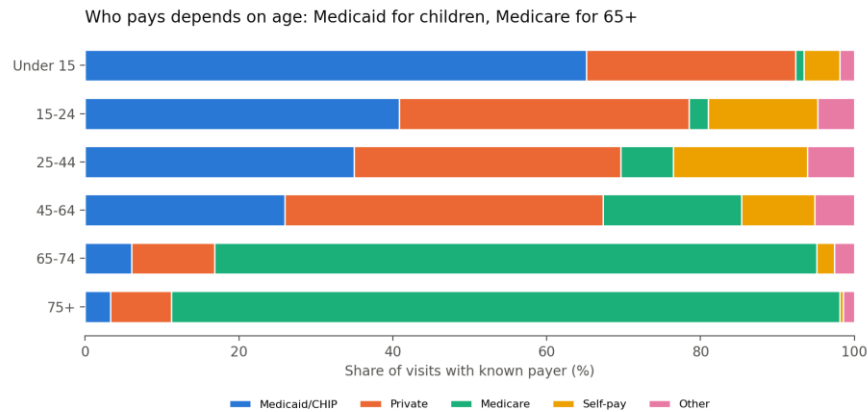


Figure 2. Primary expected payer by age group (weighted, known payer)

5. Clinical profile: chronic disease, injuries, diagnostics, medications (Q6-Q9)

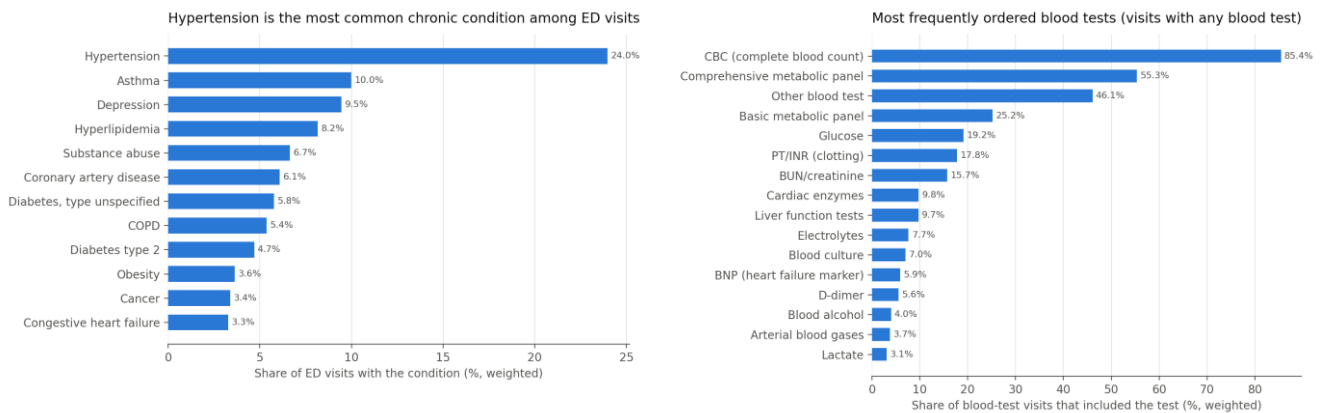


Figure 3. Most common chronic conditions (left) and blood tests ordered at blood-test visits (right), weighted

Chronic conditions (Q6). Excluding the 1.6% of visits where the section was blank, **47.6% of visits involve at least one chronic condition** (48.5% unweighted) and 14.9% involve three or more. TOTCHRON matches the 22 individual checkboxes on 100% of visits. **Hypertension is most prevalent (24.0%)**, followed by asthma (10.0%) and depression (9.5%). Diabetes type 2 alone is 4.7% and obesity 3.6%, but diabetes is split across three checkboxes: combined, **any diabetes covers 11.1%**, making it the second most common condition. Chronic burden rises from 13% of under-15 visits to 90% of visits by patients 75+. EDs are therefore a front line for chronic disease: many visits are flare-ups of conditions that primary care could manage. That argues for ED-based care coordination, follow-up referrals and medication reconciliation.

Injuries (Q7). **33.1% of visits** (weighted) relate to an injury (29.5%), an overdose or poisoning (1.4%) or an adverse effect of medical care (2.2%). Intent is never recorded for adverse effects (the question is only asked for injuries and overdoses), and it's blank for 21% of injuries. **Among visits with known intent, 7.3% are intentional and 92.7% unintentional.** Intentional shares are highest for overdoses (13.2%) and patients aged 15-24 (12.1%).

Diagnostics (Q8). **47.0% of visits include imaging** (X-ray 33.7%, CT 16.5%, ultrasound 4.4%, MRI 0.7%), rising from 29% for children to 70% for patients 75+, and from 44% of non-injury visits to 54% of injury visits. 42.4% of visits include a blood test. At those visits, **CBC is by far the most common (85.4%)**, followed by the comprehensive (55.3%) and basic (25.2%) metabolic panels, with 3.2 different tests per visit on average. Standalone glucose, BUN/creatinine and electrolytes are low because the panels already include them.

Medications (Q9). The weighted **mean is 2.49 medications per visit and the median is 2**. **20.9% of visits involve none, 41.9% involve 1-2 and 37.2% involve 3 or more**, exactly matching NCHS (Table 1). On average 1.62 are given in the ED and 1.08 prescribed at discharge. Treatment intensity tracks complexity: 2.0 medications with no chronic condition vs 3.8 with three or more, peaking at 3.1 for ages 45-64. More than a third of visits are pharmacologically complex, which raises medication-safety and pharmacy-staffing needs.

6. Correlations and outcomes (Q10)

We built about 20 cleaned variables (codes set to missing, wait and length of visit winsorized) and computed **weighted** Pearson correlations using rows where both variables are present. Triage level runs from 1 = immediate to 5 = nonurgent, so a negative r means "more for urgent patients".

Table 3. Selected weighted correlations

Pair	Weighted r	Reading
Age - Medicare	0.61	Mechanical: Medicare eligibility starts at 65
Age - # chronic conditions	0.55	Chronic burden accumulates with age
# blood tests - admitted	0.37	Workup intensity signals severity
Triage level - # blood tests	-0.35	Urgent patients get more tests
# chronic conditions - admitted	0.33	Complex patients are admitted more
Length of visit - # blood tests	0.31	Diagnostics lengthen stays
Wait - length of visit	0.28	Front-end delays carry through
Arrived by ambulance - admitted	0.24	Ambulance arrivals are sicker
Sex, ethnicity, pain - any outcome	about 0	No meaningful association

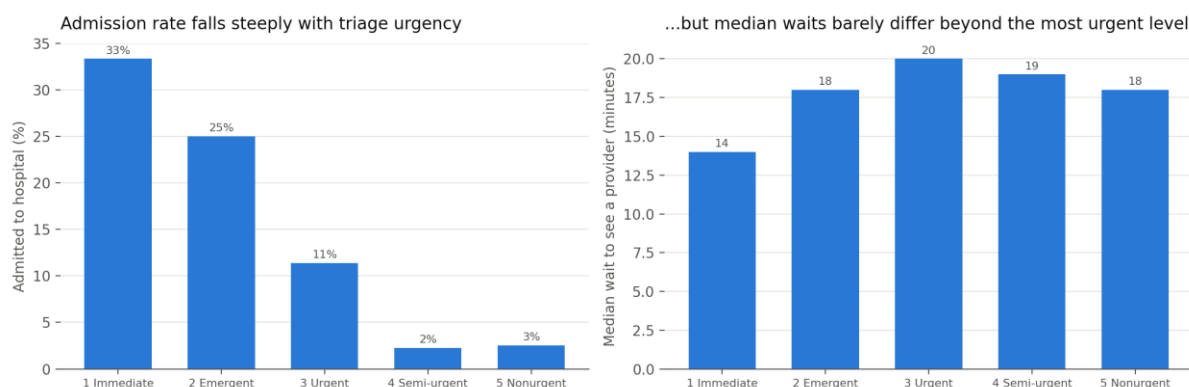


Figure 4. Admission rate (left) and median wait (right) by triage level, weighted

Outcomes. Deaths are too rare for correlations to capture: only 28 sampled visits died in the ED (0.10% weighted), 7 were dead on arrival (0.04%), and 40 of 1,752 admitted patients died in hospital (2.6%). Their correlations are near zero even where the relationship is strong, so we compare rates across groups instead. By triage level, the admission rate falls from **33% (immediate) to 2-3% (semi- and nonurgent)**, **6.4% of immediate patients die in the ED**, and 41% of them arrive by ambulance. By age, admission rises from 2.5% (under 15) to 29.1% (75+). The operational surprise: **median waits are 14 minutes for immediate patients but 18-20 minutes for every other level**, so triage fast-tracks only the sickest, and emergent patients wait as long as nonurgent ones. These are associations, not causal effects: age simultaneously drives chronic burden, payer and treatment intensity.

7. Implications for hospital management, and limitations

- **Capacity:** staff to a 10am-8pm plateau with a Monday premium. Throughput time (a median 2.6-hour stay with a long tail) is as important a capacity lever as arrival volume.
- **Triage and flow:** emergent (level 2) patients wait as long as nonurgent ones. A fast track for levels 4-5 could free provider time for level 2-3 patients.
- **Finance:** 55% public payers and 11% uninsured or charity. Accurate payer capture (10.8% unknown) and Medicaid contracting matter for margins.
- **Chronic care:** nearly half of visits involve chronic disease. Care coordination and follow-up could shift avoidable visits to cheaper settings.
- **Limitations:** these are weighted point estimates without survey-design standard errors (CSTRATM/CPSUM; NCHS advises testing at the 1% level). Wait-time missingness is not random. Chronic conditions such as obesity are likely under-recorded, and the 2015 form changes limit comparisons with earlier years.

AI use. As the course requires, we used AI throughout. The analysis code was written with Claude (Anthropic) through Claude Code, directed by the group question by question. We ran every step ourselves on the Yale HPC cluster, reviewed each output before moving on, and verified results against published NCHS figures (20 of 20 checks pass). This write-up and the slides were drafted with AI assistance from those verified outputs, then reviewed and edited by the group.